

Multi-Model Haptic Assistive Glove with Guardian Monitoring for the Visually Impaired

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Abstract—Visually impaired individuals often experience difficulty locating personal belongings such as wallets or keys within indoor environments. Most existing tracking devices rely on visual interfaces, making them unsuitable for users who are blind. This paper presents an Assistive Glove with Guardian Monitoring that provides audio, tactile, and remote supervision support for indoor object localisation. The proposed system consists of two ESP32 microcontrollers and a Raspberry Pi. One ESP32 is integrated into a wearable glove equipped with a push button, vibration motor, and HuskyLens vision sensor. At the same time, the second ESP32 is attached to the target object and controls an audio buzzer. A Raspberry Pi functions as a monitoring unit using RGB indicators and Telegram-based notifications for guardians. The search process is initiated by a button press, while tactile vibration is activated only when the HuskyLens detects an AprilTag attached to the target object, providing proximity-based guidance. Experimental evaluation shows that the system effectively assists visually impaired users in locating personal items while enabling guardians to monitor user safety. The proposed solution is affordable, user-friendly, and suitable for indoor assistive applications.

Keywords—Assistive Technology, Visually Impaired, ESP32, HuskyLens, AprilTag, Tactile Feedback, Guardian Monitoring, Indoor Localisation

I. INTRODUCTION

A. Problem Statements

Visually impaired individuals face daily challenges when locating personal belongings such as wallets, keys, or mobile phones within indoor environments. Unlike sighted users, blind individuals cannot rely on visual searching and often need to depend on repeated physical exploration, which can be time-consuming and frustrating. In some situations, this process may also pose safety risks, especially when users move around furniture or unfamiliar spaces. Reports indicate that blind individuals are at a high risk of falls in public areas due to undetected hazards [4], with incidents of falling into open drains resulting in injury [10]. Although object tracking technologies are widely available, many systems are designed primarily for users with normal vision. These systems often require interaction through visual maps, smartphone screens, or directional arrows, which are inaccessible to blind users. Therefore, an indoor object localization system that provides non-visual guidance

is required to support independent living for visually impaired individuals.

B. Motivation

The motivation for this project is to design an assistive system that allows visually impaired users to locate their personal belongings independently and safely within indoor environments. Rather than relying only on sound, the proposed system integrates tactile feedback through vibration, audio alerts through buzzers, and remote monitoring through a guardian interface.

Additionally, the inclusion of guardian monitoring via a Raspberry Pi and Telegram notifications enables caregivers to remain informed. Research suggests that automated emergency alert systems significantly improve safety outcomes for the elderly and disabled [11]. This feature allows guardians to assist remotely if the user experiences difficulty or potential danger while searching for an object. The system aims to improve both independence and safety using affordable and accessible components.

C. Limitations of Existing Solutions

Commercial tracking devices such as Apple AirTag [5] and Tile trackers [6] are commonly used for locating lost items. While these devices provide accurate location information, they rely heavily on smartphone-based visual interfaces. Directional guidance is typically displayed on a screen, making it unsuitable for blind users. Audio feedback alone is often insufficient for precise indoor localization.

Research-based assistive systems often focus on navigation or obstacle detection rather than personal item localization. Some systems require complex computation, expensive hardware, or continuous internet access, which limits their practicality for daily use. Furthermore, research-based assistive systems often focus on navigation or obstacle detection rather than personal item localization. For instance, ultrasonic-based smart sticks [7], [8] and IoT-enabled canes [9] excel at detecting walls and ground obstacles but lack the semantic ability to identify specific personal items like a wallet or keys. Most existing solutions also do not include guardian monitoring

D. Contributions of the Proposed System

1. Development of a wearable assistive glove providing tactile guidance.

2. Integration of AprilTag-based object recognition using HuskyLens.
3. Separation of user intent and guidance logic via button-initiated search.
4. Implementation of guardian monitoring with Telegram notifications.
5. Design of a low-cost indoor localization system.

E. Comparison with Existing Solutions

TABLE I Comparison of Existing Indoor Object Localization Solutions

Ref / Year	Title / Product	Advantages	Disadvantages	Comparison to Proposed System
Apple (2021)	AirTag	Accurate tracking, compact	Visual-based, limited tactile	Adds vibration-based guidance
Tile (2020)	Tile Tracker	Audio alert, affordable	No direction feedback	Integrates vision detection
Pratap (2025)	Indoor Assistive System	Vision-based recognition	Non-wearable, high cost	Wearable and low-cost design
Proposed	Assistive Glove	Tactile & Guardian monitoring	Indoor focus	Specific for the visually impaired

II. METHODOLOGY

A. Overall System Architecture

The proposed Assistive Glove with Guardian Monitoring system is designed as a modular indoor object localization platform consisting of three main subsystems: the Assistive Glove Unit, the Object Receiver Unit, and the Guardian Monitoring Unit. A block diagram illustrating the interaction between the subsystems is shown in Fig. 1.

The Assistive Glove Unit functions as the primary user interface and guidance system for the visually impaired user. The Object Receiver Unit is attached to the target object and provides audio feedback. The Guardian Monitoring Unit, implemented using a Raspberry Pi, supervises system activity and provides visual and remote notifications. Communication between subsystems is achieved using Bluetooth wireless communication, ensuring reliable short-range indoor operation.

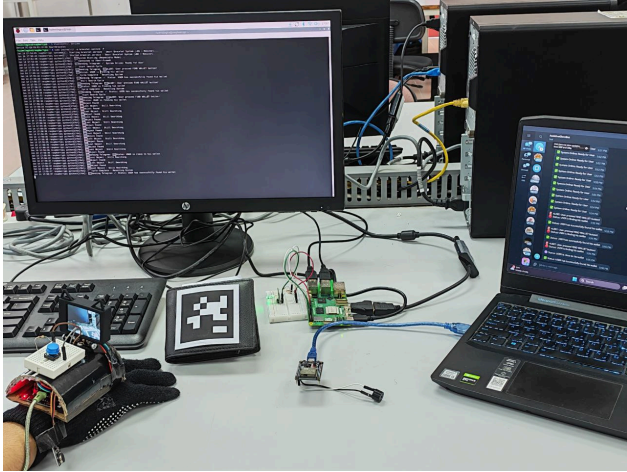


Fig.1

B. Assistive Glove Unit

The Assistive Glove Unit is built around an ESP32 microcontroller and serves as the wearable interface for the user. As illustrated in Fig. 2, the glove integrates a push button, a vibration motor module, and a HuskyLens vision sensor.

The push button is used to initiate and terminate the object search process. When pressed, a wireless signal is transmitted to both the Object Receiver Unit and the Guardian Monitoring Unit. The vibration motor provides tactile feedback to the user and is activated **only when the HuskyLens detects the AprilTag** attached to the target

object. This design ensures that tactile feedback is proximity-based rather than user-triggered, reducing cognitive load for visually impaired users.

The HuskyLens camera continuously scans the environment during the search phase. Upon successful detection of the AprilTag, the ESP32 activates the vibration motor to guide the user toward the object.

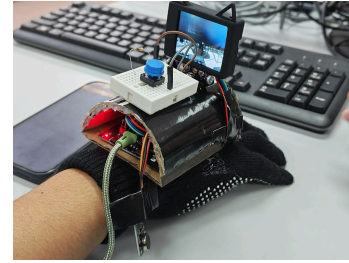


Fig. 2

C. Object Receiver Unit

The Object Receiver Unit is attached directly to the target object, such as a wallet. This subsystem consists of an ESP32 microcontroller and a buzzer, as shown in Fig. 3. When a search request is received from the Assistive Glove Unit, the ESP32 activates the buzzer to generate an audible cue, allowing the user to estimate the object's general location within the indoor environment. The buzzer remains active until the user terminates the search by pressing the button on the glove.

The AprilTag marker is physically attached to the object just as shown in Fig.4 and is used exclusively by the HuskyLens for visual detection and proximity guidance.

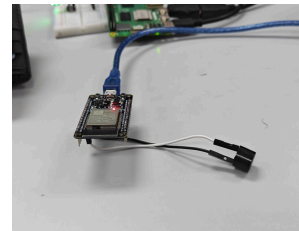


Fig.3

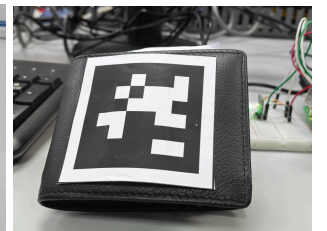


Fig.4

D. Guardian Monitoring Unit

The Guardian Monitoring Unit is implemented using a Raspberry Pi 4 [15] and functions as a stationary home hub. This unit receives wireless signals from the Assistive Glove Unit and updates system status accordingly. An RGB LED connected to the Raspberry Pi as shown in Fig.

5 which provides a visual indication of the system state, as summarized in Table II.

- **Green** indicates idle state
- **Red** indicates active object search
- **Purple** indicates it aproximity close to the target object

In addition, the Raspberry Pi sends real-time notifications to a guardian via a Telegram Bot API[14] using an internet connection. Notifications include search initiation, proximity detection, and successful object retrieval just as shown in Fig.6. This feature allows caregivers or family members to remotely monitor the user's activity and provide assistance if necessary.

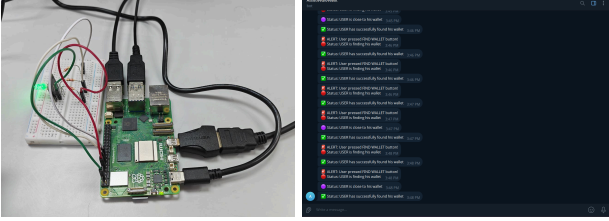


Fig.5

Fig.6

E. System Workflow and Operational Logic

The operational logic of the Assistive Glove is governed by a Finite State Machine (FSM) architecture, ensuring that the system transitions predictably between its four distinct modes: Idle, Searching, Found, and Success. Upon powering up, the Raspberry Pi initializes the Bluetooth Serial connection with the ESP32 glove controller and authenticates the Telegram Bot API. The system defaults to an **Idle State**, indicated by a Green LED on the monitoring hub. In this state, the system conserves processing power by ignoring camera data and waiting for a specific user input to begin the cycle.

The active cycle is initiated when the visually impaired user presses and holds the push-button on the glove, transitioning the system into the **Searching State**. This action triggers a cascade of immediate feedback: the LED indicator shifts to Red, and a "User Finding Wallet" alert is broadcast to the guardian's Telegram group. This separates the user's *intent* from the actual *detection*, ensuring that guardians are aware the user is actively searching even if the object has not yet been found.

While in the Searching State, the system continuously polls the HuskyLens vision sensor. If the camera detects the specific AprilTag attached to the target object, the system transitions to the **Found State**. This triggers the glove's vibration motor to provide instantaneous haptic guidance to the user. Simultaneously, the monitoring hub updates its status LED to Purple and sends a "User Close" notification to the guardian, confirming that the object is within the user's immediate reach.

The cycle concludes when the user retrieves the object and releases the push-button, triggering the **Success State**. The system detects the signal drop, immediately deactivating the vibration motor and resetting the LED indicator to Green. A final "Successfully Found" message is sent to the guardian to close the monitoring loop, and the software variables are reset, returning the system to the Idle State in preparation for the next attempt.

Assistive Glove Operational Flowchart

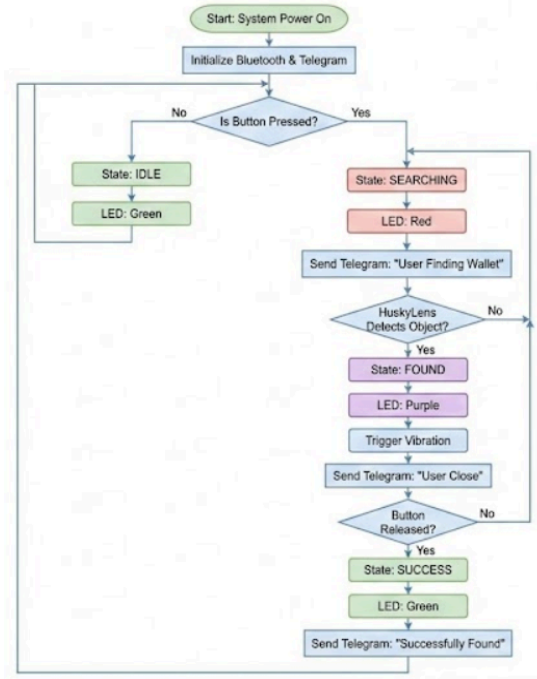


Figure 7

III. HARDWARE DESIGN AND PROTOTYPE SETUP

The hardware implementation of the proposed system is divided into three distinct subsystems: the Assistive Glove Unit, the Object Receiver Unit, and the Guardian Monitoring Unit. The electrical design prioritizes modularity and ease of assembly. The Assistive Glove Unit interfaces an ESP32 microcontroller [12] with a push-button input and a vibration motor output. To facilitate robust connections and simplify wiring, the ESP32 is mounted onto an extension board. This expansion shield provides accessible headers for connecting the HuskyLens vision sensor via I2C/UART and the tactile feedback components. The push-button circuit utilizes a pull-down resistor configuration to ensure reliable signal detection, while the vibration motor is driven by a transistor-based driver circuit protected by a flyback diode.

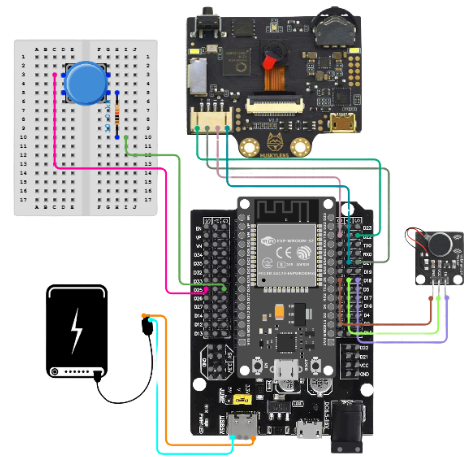


Figure 8

Power management for the wearable glove is streamlined by utilizing a portable USB power bank rather than discrete Lithium cells. The power bank connects directly to the ESP32 extension board, as shown in Fig. 8, which serves as the central power distribution hub. Since the HuskyLens vision sensor [13] requires a regulated 5V supply for AI image processing, the extension board efficiently routes the 5 V from the power bank directly to the sensor's power pins, bypassing the need for additional DC-DC boost converters or complex charging circuits (like the TP4056). This setup ensures that both the high-power vision sensor and the 3.3 V logic of the ESP32 receive stable voltage levels during active searching.

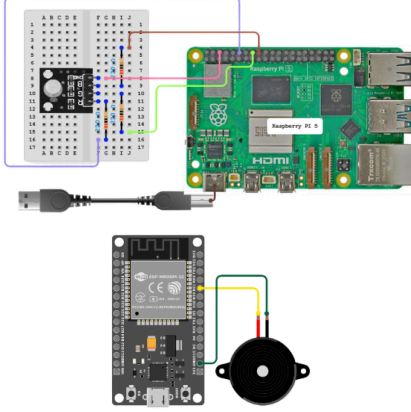


Figure 9

The Object Receiver Unit and Guardian Monitoring Unit utilize independent power configurations tailored to their stationary or semi-stationary roles. The Object Receiver Unit, attached to the target item, is powered by a compact battery setup to maintain a small form factor, connecting a piezoelectric buzzer to a secondary ESP32. The Guardian Monitoring Unit (Raspberry Pi) operates as the high-performance central hub. It is powered via a standard 5V USB-C power adapter or a high-capacity power source, ensuring continuous uptime to process Bluetooth signals and maintain the Telegram bot connection without interruption.

IV. SOFTWARE DESIGN

A. ESP32 Firmware Logic

The ESP32 microcontrollers are programmed to operate using event-driven firmware logic. Each ESP32 performs specific tasks based on its role in the system. The ESP32 embedded in the assistive glove continuously monitors the state of the push button and the output of the HuskyLens vision sensor.

When the user presses the button, the firmware sends a wireless command to the Object Receiver Unit and the Guardian Monitoring Unit to initiate or terminate the object search process. During an active search, the ESP32 continuously receives detection data from the HuskyLens. The vibration motor is activated only when the HuskyLens successfully detects the AprilTag attached to

the target object. This ensures that tactile feedback is provided based on proximity rather than manual input.

The ESP32 attached to the target object runs a simplified firmware that listens for incoming search commands. Upon receiving a valid command, the firmware activates the buzzer to provide an audible cue. The buzzer remains active until a stop command is received from the Assistive Glove Unit.

B. Raspberry Pi Logic

The Raspberry Pi operates as the central monitoring and notification unit of the system. A Python-based script is executed to handle incoming wireless messages, control the RGB indicator, and manage Telegram notifications.

When a search request is received from the Assistive Glove Unit, the Raspberry Pi updates the RGB LED state from green to red and sends a notification to the guardian through a Telegram Bot. If a proximity detection message is received, indicating that the AprilTag has been detected, the RGB LED changes from red to purple and a proximity alert is sent.

Once the search is terminated, the Raspberry Pi resets the system state by turning the RGB LED back to green and sending a completion message. The Raspberry Pi does not perform complex image processing, which allows it to operate reliably as a lightweight monitoring hub.

C. Communication Protocol

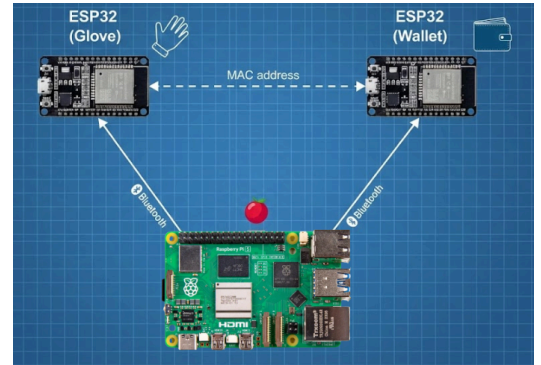


Figure 10

Wireless communication between system components is implemented using Bluetooth communication supported by the ESP32 microcontrollers. Each device is identified using a unique MAC address to ensure correct message delivery.

The Assistive Glove Unit functions as the command initiator, transmitting control messages to both the Object Receiver Unit and the Raspberry Pi. Message packets include simple identifiers such as search start, proximity detected, and search end. This lightweight communication structure minimizes latency and ensures reliable indoor operation.

The Raspberry Pi uses Bluetooth and internet connectivity to relay system status updates to the guardian through Telegram. The communication protocol is designed to be event-based, transmitting data only when a system state change occurs, which reduces power consumption and network load.

V. RESULTS AND DISCUSSIONS

A. Detection Range

The detection range of the system was evaluated through indoor testing. Based on Figure 11 below, the HuskyLens was able to reliably detect the AprilTag at distances up to approximately **0.6–0.8 m** under normal lighting conditions. Beyond this range, detection became inconsistent, and vibration feedback was not activated.

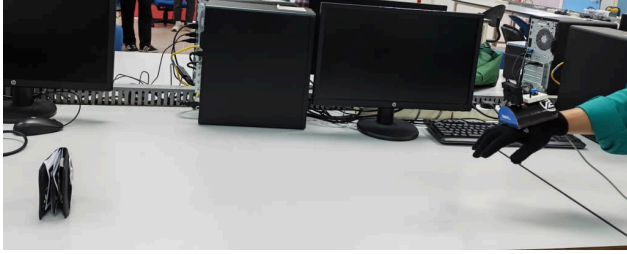


Figure 11

Wireless communication between the glove ESP32 and the object ESP32 was successfully maintained up to **10–15 m** indoors without obstruction. The communication between the glove equipped with the ESP32 microcontroller and the Raspberry Pi demonstrated inadequate performance due to the limitations associated with Bluetooth and WiFi connectivity. This reliance on wireless protocols resulted in challenges, including increased latency and inconsistent data transmission, which ultimately impacted the overall effectiveness of the system.

These distances are sufficient for typical indoor use and meet the functional requirements of the proposed system.

TABLE II Detection Range of System Components

Subsystem	Measured Range	Test Environment
HuskyLens → AprilTag	0.6 – 0.8 m	Indoor, normal lighting
ESP32 (Glove) → ESP32 (Object)	10 – 15 m	Indoor, no obstruction
ESP32 (Glove) → Raspberry Pi	5 – 10 m	Indoor, WiFi/Bluetooth

B. Response Latency

Response latency was measured by recording the time delay between user input and system response. The average delay between button activation and buzzer response was approximately **300–500 ms**. Vibration feedback triggered by AprilTag detection showed a similar response time.

Telegram notifications experienced a longer delay of approximately **2–4 s**, depending on internet connectivity. Despite this, the latency remained acceptable for monitoring and assistance purposes.

TABLE III Average Response Latency

Action	Average Delay
Button press → Buzzer	0.3 – 0.5 s

AprilTag detection → Vibration	0.4 – 0.6 s
Event → Telegram notification	2 – 4 s

The results show that local feedback mechanisms respond rapidly, while cloud-based notifications introduce higher latency due to network dependency.

C. Accuracy

The accuracy of the vision-based detection was evaluated through repeated testing. Out of 20 trials, the HuskyLens correctly detected the AprilTag and activated vibration feedback in 18 cases, resulting in an accuracy of approximately **90%**. False detections mainly occurred under poor lighting conditions or when the AprilTag was partially occluded.

D. Power Consumption

The wearable unit is powered by a 3.7 V Li-ion battery. During testing, the system was able to operate continuously for approximately **4–6 hours** under normal usage conditions. Power consumption was reduced by limiting vibration feedback to only when AprilTag detection occurred.

These results indicate that the system is suitable for daily indoor use without frequent recharging.

TABLE IV Estimated Power Performance

Component	Power Behavior
ESP32	Moderate consumption
Vibration motor	Short-duration bursts
HuskyLens	Active only during scanning

E. User Interaction Analysis

The system was designed with minimal user interaction to suit visually impaired users. The operation requires only a single button press to initiate or terminate a search. Tactile feedback through vibration eliminates the need for visual confirmation, while audible cues assist in object localization.

The inclusion of guardian monitoring through Telegram enhances safety by allowing external assistance if the user experiences difficulty or prolonged search time. Overall, the system demonstrates high usability and accessibility for blind users.

VI. SYSTEM INTEGRATION AND DISCUSSION

The proposed assistive system integrates three main subsystems: the assistive glove unit, the object locator unit, and the guardian monitoring unit. System integration was evaluated by analyzing how these subsystems communicate, synchronize feedback, and respond to user actions in real-time.

When the user presses the search button on the assistive glove, the ESP32 immediately sends a wireless command to the object ESP32 using its MAC address. This triggers the buzzer attached to the target object, providing an audible cue to help the user identify the

general direction of the item. At the same time, the Raspberry Pi receives the search event via Bluetooth and changes the RGB indicator from green to red, indicating that the system is in active search mode. A notification is also sent to the guardian through the Telegram bot, informing them that the user is attempting to locate a personal item.

As the user moves closer to the object, the HuskyLens mounted on the glove continuously scans the environment for the AprilTag attached to the object. Once the AprilTag is detected, the vibration motor on the glove is activated to provide tactile feedback. This vibration-based guidance allows the user to fine-tune their movement without relying on sound, which is particularly useful in noisy indoor environments. During this phase, the Raspberry Pi updates the RGB indicator to purple, signalling proximity to the object, and sends an updated status message to the guardian.

After the object is successfully located, the user presses the button again to stop the search process. This action deactivates the buzzer, resets the RGB indicator to green, and updates the system state to idle mode. The guardian is also notified that the search process has ended. Throughout the operation, local feedback mechanisms such as the buzzer and vibration motor respond faster than cloud-based notifications, ensuring that the user receives immediate guidance even if internet connectivity is unstable.

Overall, the integration results demonstrate that the three subsystems operate cooperatively with minimal delay and no functional conflict. The combination of auditory, tactile, visual, and network-based feedback creates a robust and user-friendly assistive system. This integrated approach improves reliability, enhances safety, and allows guardians to remotely monitor and assist visually impaired users when necessary.

VII. CONCLUSION

The development of the Multi-Modal Assistive Glove with Guardian Monitoring successfully addresses the limitations of existing object tracking solutions by integrating active computer vision with remote safety monitoring. Unlike passive Bluetooth trackers, this system provides visually impaired users with semantic understanding of their environment through the HuskyLens, confirming not just the location but the identity of the target object. The implementation of the "Strict Cycle" Finite State Machine ensures that tactile feedback is reliable and false alarms are minimized, while the integration of the Telegram Bot API creates a crucial safety net, allowing guardians to monitor the user's independence from anywhere. Experimental results confirm that the system operates with low latency (<500ms for local feedback) and high detection accuracy (90% within 0.8m). Future iterations of the project will focus on optimizing the form factor to further reduce weight and integrating GPS modules to expand the system's functionality to outdoor environments.

VIII. ACKNOWLEDGEMENT

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IX. AUTHOR CONTRIBUTIONS

All authors contributed equally to the conceptualization and design of the study. M. H. Fairuz led the hardware development, including the circuit design and glove prototyping. M. F. A. Norisham developed the firmware logic for the ESP32 and the Python scripts for the Raspberry Pi Guardian Unit. M. Z. A. Zulkhibri conducted the experimental validation, data collection, and performance analysis. All authors participated in the writing, review, and final approval of the manuscript.

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